

MATH3018 Assignment 3, due on Feb 21

1. Suppose that the random vector $X = (X_1, X_2)^T$ has a bivariate normal distribution with $\text{Var}(X_1) = \text{Var}(X_2)$. Show that $X_1 + X_2$ and $X_1 - X_2$ are independent random variables. [2.5pts]

Solution: Let

$$\Sigma = \text{Cov}(X) = \begin{pmatrix} \sigma_{11} & \sigma_{12} \\ \sigma_{12} & \sigma_{11} \end{pmatrix}, Y = \begin{pmatrix} X_1 + X_2 \\ X_1 - X_2 \end{pmatrix}, A = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}.$$

i.e. $Y = AX$. Therefore Y follows a bivariate normal distribution,

with

$$\text{Cov}(Y) = A\Sigma A^T = \begin{pmatrix} 2\sigma_{11} + 2\sigma_{12} & 0 \\ 0 & 2\sigma_{11} - 2\sigma_{12} \end{pmatrix}.$$

This implies that $Y_1 = X_1 + X_2$ and $Y_2 = X_1 - X_2$ are independent.

2. Suppose $X = (X_1, X_2, X_3)^T$ follows $N_3(\mu, \Sigma)$, where

$$\mu = \begin{pmatrix} -3 \\ 1 \\ 4 \end{pmatrix}, \Sigma = \begin{pmatrix} 1 & -2 & 0 \\ -2 & 5 & 0 \\ 0 & 0 & 2 \end{pmatrix}.$$

Justify whether each of the following pairs is independent or not [0.5pt each]:

- (a) X_1 and X_2 .
- (b) X_2 and X_3 .
- (c) $(X_1, X_2)^T$ and X_3 .
- (d) $X_1 + X_2$ and X_3 .
- (e) X_2 and $-2X_1 + X_2 - X_3$.

Solution

- (a) $\text{Cov}(X_1, X_2) = -2$, not independent.
- (b) $\text{Cov}(X_2, X_3) = 0$, independent.
- (c) $\text{Cov}\left((X_1, X_2)^T, X_3\right) = (0, 0)^T$, independent.
- (d) $\text{Cov}(X_1 + X_2, X_3) = 0$, independent.
- (e) $\text{Cov}(X_2, -2X_1 + X_2 - X_3) = 9$, not independent.

3. Let $\mathbf{X} = \begin{bmatrix} \mathbf{X}_1 \\ \mathbf{X}_2 \end{bmatrix}$ be distributed as $N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ with $\boldsymbol{\mu} = \begin{bmatrix} \boldsymbol{\mu}_1 \\ \boldsymbol{\mu}_2 \end{bmatrix}$, $\boldsymbol{\Sigma} = \begin{bmatrix} \boldsymbol{\Sigma}_{11} & \boldsymbol{\Sigma}_{12} \\ \boldsymbol{\Sigma}_{21} & \boldsymbol{\Sigma}_{22} \end{bmatrix}$, and $|\boldsymbol{\Sigma}_{22}| > 0$. Then the conditional distribution of $\mathbf{X}_1$, given that $\mathbf{X}_2 = \mathbf{x}_2$, is normal and has mean $\boldsymbol{\mu}_1 + \boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}(\mathbf{x}_2 - \boldsymbol{\mu}_2)$ and covariance $\boldsymbol{\Sigma}_{11} - \boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}\boldsymbol{\Sigma}_{21}$

(Hint: consider taking the linear transformation $\mathbf{A}\mathbf{X}$ where $\mathbf{A} = \begin{pmatrix} \mathbf{I}_q & -\Sigma_{12}\Sigma_{22}^{-1} \\ 0_{(p-q)\times q} & \mathbf{I}_{(p-q)} \end{pmatrix}$, and where q denotes the dimension of $\mathbf{X}_1$.) [3pts]

Solution

$$AX = \begin{pmatrix} X_1 - \Sigma_{12}\Sigma_{22}^{-1}X_2 \\ X_2 \end{pmatrix} \sim N_p(A\mu, A\Sigma A^T), \text{ where } A\mu = \begin{pmatrix} \mu_1 - \Sigma_{12}\Sigma_{22}^{-1}\mu_2 \\ \mu_2 \end{pmatrix} \text{ and}$$

$$A\Sigma A^T = \begin{pmatrix} \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21} & 0_{q\times(p-q)} \\ 0_{(p-q)\times q} & \Sigma_{22} \end{pmatrix}$$

indicating that $X_1 - \Sigma_{12}\Sigma_{22}^{-1}X_2$ and X_2 are independent. Therefore,

$$\begin{aligned} & \text{conditional distribution of } X_1 - \Sigma_{12}\Sigma_{22}^{-1}X_2 \text{ given } X_2 = x_2 \\ &= \text{unconditional distribution of } X_1 - \Sigma_{12}\Sigma_{22}^{-1}X_2 \\ &= N_q(\mu_1 - \Sigma_{12}\Sigma_{22}^{-1}\mu_2, \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21}) \end{aligned}$$

In addition, when given $X_2 = x_2$, $\Sigma_{12}\Sigma_{22}^{-1}X_2$ is just a constant vector $\Sigma_{12}\Sigma_{22}^{-1}x_2$. In consequence, conditional distribution of X_1 given $X_2 = x_2$ is $N_q(\mu_1 + \Sigma_{12}\Sigma_{22}^{-1}(x_2 - \mu_2), \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21})$.

4. Suppose $Y = (Y_1, Y_2, Y_3, Y_4)^T$ follows $N_4(\mu, \Sigma)$, where

$$\mu = \begin{pmatrix} 1 \\ 2 \\ 3 \\ -2 \end{pmatrix}, \Sigma = \begin{pmatrix} 4 & 2 & -1 & 2 \\ 2 & 6 & 3 & -2 \\ -1 & 3 & 5 & -4 \\ 2 & -2 & -4 & 4 \end{pmatrix}.$$

Find the following [1pt each]:

- (a) The marginal distribution of $(Y_1, Y_3)^T$.
- (b) The marginal distribution of Y_2 .
- (c) The distribution of $Z = Y_1 + 2Y_2 - Y_3 + 3Y_4$.
- (d) The conditional distribution of $X_1 = (Y_1, Y_2)^T$ given that $X_2 = (Y_3, Y_4)^T = (y_3, y_4)^T = x_2$.

Solution

(a)

$$\begin{pmatrix} Y_1 \\ Y_3 \end{pmatrix} \sim N_2\left(\begin{pmatrix} \mu_1 \\ \mu_3 \end{pmatrix}, \begin{pmatrix} \sigma_{11} & \sigma_{13} \\ \sigma_{31} & \sigma_{33} \end{pmatrix}\right) = N_2\left(\begin{pmatrix} 1 \\ 3 \end{pmatrix}, \begin{pmatrix} 4 & -1 \\ -1 & 5 \end{pmatrix}\right).$$

(b) $Y_2 \sim N(\mu_2, \sigma_{22}) = N(2, 6)$.

(c) Let $a = (1, 2, -1, 3)^T$ i.e. $Z = a^T Y$. Then $Z \sim N(a^T \mu, a^T \Sigma a) = N(-4, 79)$.

(d) Let

$$\mu_1 = \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \mu_2 = \begin{pmatrix} 3 \\ -2 \end{pmatrix}, \Sigma_{11} = \begin{pmatrix} 4 & 2 \\ 2 & 6 \end{pmatrix}, \Sigma_{22} = \begin{pmatrix} 5 & -4 \\ -4 & 4 \end{pmatrix}, \Sigma_{12} = \Sigma_{21}^T = \begin{pmatrix} -1 & 2 \\ 3 & -2 \end{pmatrix}.$$

Then

$$X_1 \left| X_2 = x_2 \sim N_2 \left(\mu_1 + \Sigma_{12} \Sigma_{22}^{-1} (x_2 - \mu_2), \Sigma_{11} - \Sigma_{12} \Sigma_{22}^{-1} \Sigma_{21} \right) = N_2 \left(\begin{pmatrix} y_3 + 1.5y_4 + 1 \\ y_3 + 0.5y_4 \end{pmatrix}, \begin{pmatrix} 2 & 2 \\ 2 & 4 \end{pmatrix} \right).$$

5. If two normal variables (or vectors) have a covariance of zero, are they necessarily to be independent? [3pts]

(Hint: Consider $X_1 \sim \mathcal{N}(0, 1)$, and $X_2 = ZX_1$ for some Z independent of X_1 and has distribution $P(Z = \pm 1) = 1/2$. Verify that X_1, X_2 are both normal, but they are not independent.)

Solution No, two normal variables with zero covariance are not necessarily independent. We can prove this with a counterexample.

Consider $X_1 \sim \mathcal{N}(0, 1)$ and $X_2 = ZX_1$ where Z is independent of X_1 and $P(Z = \pm 1) = \frac{1}{2}$. First, let's show X_2 is normally distributed: For any $t \in \mathbb{R}$:

$$\begin{aligned} P(X_2 \leq t) &= P(ZX_1 \leq t) \\ &= P(ZX_1 \leq t | Z = 1)P(Z = 1) + P(ZX_1 \leq t | Z = -1)P(Z = -1) \\ &= \frac{1}{2}P(X_1 \leq t) + \frac{1}{2}P(-X_1 \leq t) \\ &= \frac{1}{2}\Phi(t) + \frac{1}{2}\Phi(t) = \Phi(t) \quad \text{where } \Phi(t) \text{ denotes the cdf of normal distribution} \end{aligned}$$

Therefore, $X_2 \sim \mathcal{N}(0, 1)$.

We can also verify $\text{Cov}(X_1, X_2) = 0$:

$$\text{Cov}(X_1, X_2) = E[X_1 X_2] - E[X_1]E[X_2] = E[X_1 Z X_1] - 0 = E[Z]E[X_1^2] = 0.$$

However, X_1 and X_2 are not independent, because if $x_1 \neq \pm x_2$, the joint density $f_{X_1, X_2}(x_1, x_2) = 0 \neq f_{X_1}(x_1)f_{X_2}(x_2)$.